

Sample Direct Examinations of a Forensic Examiner

Direct Examination: Pretrial Evidentiary Hearing

A. Preface

Challenges to the qualifications of the computer forensic examiner or electronic evidence, which include the tools and/or methodologies used in the forensic examination process, should be addressed outside the presence of the jury utilizing a pretrial hearing pursuant to Evidence Code section 402. Many prosecutors are in agreement that contested testimony on complex technical issues should be avoided in the presence of the jury, but rather dealt with in a separate hearing beforehand. Topics addressed here include: an overview of computer forensics; the acquisition process; the authentication process; the recovery process; and addressing *Kelly-Frye* factors, among others.

EnCase is a common industry standard software application used in forensic examination analysis. It was designed specifically as a tool for computer forensic investigations. This document contains a “mock trial” direct examination where the foundation was established utilizing EnCase in an Evidence Code section 402 hearing. The example used here contains more detail than what would normally be presented during a direct examination in the context of a court trial or hearing, however, much of the information may be found useful for re-direct examination.

B. Background

[After stating name for the record]

Q: Sir, are you a Senior Special Agent for the United States Customs Service?

A: Yes I am.

Q: And do you have any specialized duties as a Customs agent?

A: I am a computer evidence examiner certified as a Seized Computer Evidence Recovery Specialist by the United States Department of the Treasury.

Q: Please tell us how long you have been a computer evidence examiner.

A: I have been a Seized Computer Evidence Recovery Specialist with Customs for eight years.

Q: Tell us about your educational background.

A: I received a Bachelor of Science degree in electrical engineering from University of _____ in 19__.¹

¹ Please note that many exceptionally qualified forensic examiners may not have attended and/or completed college.

Q: And could you briefly describe your training for the handling and examination of computer evidence?

A: In 19__ I received three-weeks of intensive training, known as Seized Computer Evidence Recovery Specialist training at the Federal Law Enforcement Training Center. In 19__ I obtained Computer Forensic Examiner Certification from the International Association of Computer Specialists, known as IACIS, after receiving two weeks of their intensive training. The next year I received Advanced Course Certification from IACIS after taking their two-week advanced training course. I have also received computer forensic training from The National Consortium for Justice Information and Statistics, known as SEARCH and have received training from Guidance Software on their EnCase computer forensic application.

Q: Are you a member of any professional organizations?

A: Yes I am.

Q: Which ones?

A: I am a member of the International Association of Computer Specialists and the High Tech Crime Investigation Association.

C. Overview of Computer Forensics

Q: You mentioned the subject of computer forensics. Can you provide an overview of what computer forensics is?

A: Computer Forensics is the acquisition, authentication and reconstruction of electronic information stored on computer media, such as hard drives, floppy disks or zip drives. A computer forensics technician is needed whenever there is evidence stored in a computer.

Q: Can you briefly tell us how a computer forensic specialist such as yourself conducts a typical investigation?

A: First, the electronic information contained on computer storage media must be acquired by making a complete physical copy of every bit of data located on computer media in a manner that does not alter that information in any way. Then the information must be authenticated in a special process that establishes that the acquired electronic information remained completely unaltered from the time the examiner acquired it. Finally, the examiner must use special software and processes to recover and reconstruct the information in its forensic state, even if such information is found in files that have been deleted by the user.

D. The Acquisition Process

Q: You described three basic steps, and I want to discuss them one at a time beginning with the acquisition process. How is digital information copied from computer media in a proper forensic manner?

A: Specialized computer forensic software, such as EnCase, utilizes a special boot process that ensures the data on the subject computer is not changed. After the boot procedure is initiated, the examiner utilizes the forensic software to create a complete forensic image copy or “exact snapshot” of a targeted piece of computer media, such as a hard drive, or external media, such as floppy or zip disks. This forensic image is a complete sector-by-sector copy of all data contained on the target media and thus all information, including available information from deleted files, is included in the forensic image created by the examiner.

E. The Authentication Process

Q: The second step you mentioned was the authentication process; please briefly describe how the acquired electronic information is authenticated and verified.

A: Computer forensic examiners rely upon software that generates a mathematical value based upon the exact content of the information contained in the forensic image copy of the seized computer media. This value is known as an MD5 hash value and is often referred to as a special type of digital signature. The same software also verifies that this value remains the same from the time it is generated. If one bit of data on the forensic image copy is subsequently altered in any way, meaning that even if a single character is changed or one space of text is added, this value changes. So if the hash value of the information contained on seized media remains the same, then it is established that the electronic data has not been altered in any way.

Q: What are the odds of two forensic images with different contents having the same hash value?

A: The odds of two computer files, including a forensic image file, with different contents having the same hash value is roughly ten raised to the 38th power. If you wrote out that number, it would be a one followed by 38 zeros. By contrast, the number one trillion written out is one followed by only twelve zeros.

F. The Recovery Process

Q: Because the third step of data recovery is complex, I am going to first ask you a few basic questions about how a computer works. First, and without being too technical, could you give us a description of how information on a hard drive is stored by the computer?

A: Yes. Basically, computer disks are storage media that are divided into concentric circles or tracks. This can be thought of as a small version of the old 78 rpm records people used to play on phonographs. The tracks are divided into sectors. Each sector has its own address, a number that is unique to that part of the disk. The operating system assigns and stores the address, so that it may retrieve all information constituting a computer file stored in a specific sector when requested by the user.

Q: How is the information recorded on the hard disk?

A: The disk is covered with a thin coat of magnetic material. When information is written to the disk, the data is recorded by magnetizing specific parts of the disk coating. The information resides there until it is overwritten.

Q: Thank you. I think we have the basic idea. I am very interested in how a computer technician can recover electronic information that has been deleted or automatically purged. Please tell us what is involved in this process.

A: When the computer user deletes electronic information, it is often assumed that the information is removed from the computer forever. That is not necessarily true. The information is still in the computer; only it is now marked by the computer to allow it to be overwritten. A general analogy would be a library card catalogue system, where the books represent files and the card catalogue represents the file directory with information as to where the files are located on the disk. When a file is deleted, its location information is removed from the card catalogue index, but the book remains on the shelf until another book randomly replaces it.

Q: To what extent can this deleted information be retrieved?

A: If the information has not yet been overwritten by other data, it is still there and can be retrieved using specialized software.

G. Authenticating the EnCase Process

Q: And what specialized software did you use for this investigation?

A: I used the computer forensic software known as EnCase.

Q: Tell us a little about the EnCase software.

A: EnCase is standard, commercially available software program that is specifically designed as a tool for computer forensic investigations. It is a fully integrated tool, meaning it performs all essential functions of a computer forensic investigation, including the imaging of a target drive, the generation of an MD5 hash of the evidentiary forensic image and the analysis of the subject evidence. The software allows for a completely non-invasive investigation in order to view all information on a computer drive, whether it is in the form of a deleted file, a non-deleted file, file fragments and even temporary or buffer files.

Q: How does the investigator use the EnCase software to recover deleted files?

A: First, EnCase creates a complete forensic image copy or “exact snapshot” of a targeted computer drive. EnCase will be able to read all existing information on that forensic image, regardless of whether the information is in the form of a deleted file that is marked by the operating system to be overwritten. Any information that has not been actually overwritten will be recovered for analysis. EnCase will organize all the files, deleted files and blocks of physical data, also known as unallocated clusters, in a convenient graphical user interface to allow the evidence to be viewed and sorted by the examiner.

Q: Does the same software perform these functions?

A: Yes. EnCase is a software process that is much more automated than other computer forensic investigation processes, as it is a fully integrated program where all the required computer forensic investigation functions are integrated into a single application in a Windows-based graphical user interface.

Q: How is the EnCase process more automated than other tools?

A: To a large extent EnCase duplicates the Windows Explorer interface and file viewing functions, with the critical added benefits of viewing deleted files and all other information on the disk that the user normally cannot see or detect without specialized software. Just as Windows Explorer presents the entire file directory and folder structure on a computer to the user in a very organized manner, EnCase will also present that information, in addition other data on the target drive in a similar manner. Other forensic software tools require a great deal of more manual steps utilizing a series of arcane DOS commands and separate tools to recreate file structures and perform separate searches on different areas of a drive.

H. Addressing *Kelly-Frye* Factors

Q: To your knowledge, is the EnCase software generally accepted in the computer forensic investigation community?

A: More than just generally accepted, EnCase is widely used in the computer forensics industry, and in my experience it is the tool of choice of the majority of computer forensic investigators in law enforcement. It is the primary computer forensic tool used by US Customs, which is my agency, and I am aware that it is the primary tool of other federal agencies, including United States Secret Service, as well as hundreds of state and local agencies. EnCase is a major part of the Seized Computer Evidence Recovery Specialist training curriculum for federal agents, and is part of the curriculum in many computer forensic training courses offered by professional organizations — most notably the annual IACIS training conference.

Q: How would one go about testing computer forensic software?

A: There are three main steps in testing computer forensic software. The first step is to generate an MD5 hash value for an image of a targeted computer drive using the forensic tool being tested and then using another standard tool to repeat the process for the same drive. The MD5 hash values generated by both tools for the same drive should be exactly the same.

The second step is to verify that whatever evidence is recovered from an evidentiary forensic image can be independently confirmed by a standard disk utility. With EnCase for instance, the program will identify the precise location on the original drive for each bit of data recovered by the examiner. With that information, the examiner can then use a disk utility such as Norton DiskEdit to independently confirm the existence and precise location of that data.

The third step is to confirm that throughout the examination process, the content on the forensic image has not been altered in any way by repeating the MD5 hash analysis of the

forensic image to verify that the MD5 hash is has not changed since the time of acquisition. These tests should be performed several times with different pieces of computer media.

Q: To what extent can EnCase be tested by a third party?

A: EnCase is commercially available and thus any examiner can purchase, use and test the program on their own. One of the advantages of the program is that all the required forensic functions are integrated into a single program with a Windows-based graphical user interface. Thus, compared to other computer forensic software, the program is easy to use.

Q: Has your agency tested the software?

A: Yes.

Q: How was it tested?

A: Before we purchased the software on a large scale, there were two computer investigation agents in my agency who conducted an extensive evaluation of the software employing the three steps I just described. I am aware that the Secret Service conducted a similar testing procedure as well. Also, since our agencies' adoption of the software we have had nearly 100 computer examination agents using the program on a daily basis in the field.

Q: What were the results of those tests?

A: By all accounts the software has met the three standards I described above.

Q: Has EnCase been subjected to any publication in the industry that you are aware of?

A: Yes, I have read various published articles in the information security and high-tech crime investigation industries that either favorably review the product or mention the product favorably. An article in the April 2001 issue of SC Magazine featured the most detailed and documented published testing results to date. The magazine gave EnCase its highest rating and noted that in its testing EnCase "outperformed all the other tools" that were tested by the magazine.

(If you have them, submit copies of published articles in the industry discussing EnCase software as an exhibit.)

Q: Thank you, Your Honor, nothing further.

Direct Examination for the Presentation of Computer Evidence Before a Jury

Many prosecutors maintain that when presenting computer evidence before a jury, the testimony should be as simple and straightforward as possible. Burdening the jury with overly technical information could prove counter-productive and may actually open the door to areas of cross-examination that the court would normally have disallowed. As such, the following direct examination is more detailed than is likely needed, but again, should provide a general resource in preparing direct examinations or for responding on re-direct. Further, there are many other foundational areas that are normally outside the scope of the EnCase process, such as establishing how an Internet chat room works, what the Windows operating system is, or establishing that the computer belonged to the defendant, which are not addressed here.

When presenting EnCase-based evidence, it is recommended that the proponent take full advantage of the EnCase process and graphical user interface by presenting screen shots of the EnCase “All Files” and other views, in order to show the full context of the electronic evidence. Printed representations of computer information or a computer program is presumed to be accurate pursuant to Evidence Code section 1552. Similarly, Evidence Code section 1553 provides that a printed representation of images stored on a video or digital medium is presumed to be accurate pursuant to Evidence Code section 1553.

When presenting evidence contained within a computer file, a screen shot of the EnCase FileView may be the best means to present a visual output. For example, if one was seeking to establish a defendant’s state of mind by presenting an electronic audit trail or connecting file date stamps, the ability to display a visual output showing various file attributes and other metadata provides a tremendous advantage to the advocate of such evidence. EnCase provides the best method to visually display all physical and logical data contained on the target drive, while showing the context of such files by displaying file metadata and other means. When providing testimony, many examiners present evidence through screenshots in a PowerPoint presentations format, or take EnCase with them into Court for a more “live” demonstration.

Please note that for sake of brevity, many of the foundational portions of the direct exam are incorporated by reference from the above section.

A. Background

[After stating name for the record]

Q: Sir, what is your current occupation?

A: I am a Senior Special Agent for the United States Customs Service.

Q: And do you have any specialized duties as a Customs agent?

A: I am a computer evidence examiner certified as a Seized Computer Evidence Recovery Specialist by the United States Department of the Treasury.

Q: What was your involvement in the investigation of this case?

A: I conducted a computer forensic investigation of the Defendant's computer to recover relevant evidence.

Q: OK, before we discuss the results of your investigation, please tell us how long you been a computer evidence examiner.

(Please refer to previous section, which is incorporated herein by reference, for foundation testimony)

* * * *

Q: Turning to the computer forensic investigation you conducted in this case, please tell us when you first came into contact with the Defendant's computer and computer disks.

A: Pursuant to a search warrant, on May 18, 2000 I seized the Defendant's computer at his home, along with seven CD-ROMs and sixteen floppy disks that were in his desk or otherwise in the vicinity of his computer.

Q: What did you do with the Defendant's computer equipment and disks after you seized them?

A: After leaving receipts for the computer and disks, I transported the items back to our lab, where I immediately proceeded to make forensic image copies of the hard drive found in the Defendant's computer. I also made forensic images of each of the CD-ROM and floppy disks. Using the EnCase software, I also generated MD5 hash values for the hard drive and for each floppy and CD-ROM disk at the same time the forensic images were made. I then logged the Defendant's computer and the floppy and CD-ROM disks as evidence and secured them into our evidence storage room.

Q: Did you then analyze the forensic images you made?

A: Yes I did.

Q: Please describe your analysis on the forensic image of the Defendants' hard drive.

B. Recovery of Hidden Files with Renamed File Extensions

A: In my analysis of the forensic image of the hard drive, I first employed an automated function of the EnCase forensic software that analyzes all the computer files on an image of a computer drive and identifies any file signature mismatches.

Q: What are file signature mismatches?

A: A file signature mismatch is a situation where the file name extension that normally identifies the file type has been renamed, usually in order to hide the true contents of a file.

Q: What is a file name extension?

A: A file name extension is an optional addition to the file name that allows a file's format to be described as part of its name so that users can quickly understand the type of file it is without having to open files on a trial and error basis. For instance, a text file will usually have a ".txt" extension and the most common type of picture file has a ".jpg" extension.

Q: How does EnCase identify file signature mismatches?

A: Most computer files containing text or graphical images have a well-defined signatures of electronic data unique to that file type. This allows file viewers to recognize the type of file, regardless of the file extension. EnCase utilizes the same process as file viewers in order to identify files that have renamed file extensions.

A: What was the result of the file mismatch analysis that you conducted in this case?

Q: The file signature mismatch analysis revealed 16 files that were renamed as text files with a "txt" extension, but were actually graphical image files that originally had a "jpg" extension until they were renamed manually. I viewed those files and upon determining that those images appeared to be child pornography, I printed out those images.

Q: Showing to you what have been pre-marked as United States exhibits 1 through 16, can you identify these exhibits?

A: Yes. These are the printouts I made of the 16 images in question that I recovered from the Defendant's hard drive.

[Exhibits are introduced into evidence.]

C. Recovery of Deleted Files

Q: Did you examine the images you made of the Defendant's floppy disks?

A: Yes I did.

Q: What did you find?

A: I found that one of the floppy disks had five files with a "jpg" extension that had been deleted, meaning that that the computer had marked the data of those files to be overwritten. However, we were able to still recover those deleted graphical image files as the data had not actually been overwritten by the computer.

Q: How did you identify those deleted files?

A: The EnCase software will automatically identify any files that are marked by the computer to be overwritten. I located and viewed those five graphical image files and upon determining that those images appeared to be child pornography, I printed out those images.

Q: Showing to you what have been pre-marked as United States exhibits 17 through 22, can you identify these exhibits?

A: Yes. These are the printouts I made of the five images that I recovered from the Defendant's reformatted floppy drive.

Q: Do these printouts accurately reflect the images that you saw on the defendant's computer?

A: Yes.

[Exhibits are introduced into evidence.]

D. Recovery of Files "Deleted" from Multiple CD-ROM Sessions

Q: Special Agent _____, did you examine the images you made of the Defendant's CD-ROM disks?

A: Yes I did.

Q: And what did you find?

A: I found that the CD-ROM disks were actually writeable, meaning that data can be written to this type of compact disk to store computer files. A special CD writing software program, such as CD Creator, is needed to write data to a writeable compact disk. One of the writeable CDs we seized from Defendant's home had multiple sessions on it. A CD session is created when the user writes any number of files to the CD. When this is done, the CD writing software will create a table of contents for that session that points the operating system to the location of the files on the CD within the session.

Q: Can files on a writeable CD be deleted?

A: Not really. Unlike a hard drive or floppy disk, data written to a CD is actually burned to the media by a small optical laser instead of being magnetized. Once data is burned to a CD, it cannot be overwritten. However, if a new session is created on the CD, the user can omit existing files from the new table of contents created for the new session. A computer operating system will only read the table of contents from the latest created session on a CD. Thus, by omitting existing files from the table of contents of a new session, those files will normally be hidden from the view of a user. Specialized software, such as EnCase will see all the sessions on a writeable compact disk and will allow the user to compare any differences in the file contents of each session.

Q: You mentioned that one of the CDs you examined had multiple sessions. What did your analysis of the multiple sessions CD reveal?

A: The CD actually had two sessions on it. Using EnCase, we discovered that the second session contained seven files with jpg extensions that were not included in the table of contents of the first session. I then examined those seven files, which turned out to be graphical images appearing to be child pornography, and printed out those images.

Q: Showing to you what have been pre-marked as United States exhibits 23 through 30, can you identify these exhibits?

A: Yes. These are the printouts I made of the seven images that I recovered from the first session of Defendant's writeable compact disk.

Q: Do these printouts accurately reflect the images that you saw on the defendant's writeable compact disk?

A: Yes.

[Exhibits are introduced into evidence.]

E. Evidence From Swap Files

Q: What else did you find in your examination of the Defendant's computer?

A: I conducted a text string search of the forensic image of the Defendant's hard drive. In the course of our investigation, we received information that the defendant had contacted a minor over the Internet who had an America Online account under the screen name Jenny86. I ran a text search by entering the keyword Jenny86, again using the EnCase software. The search registered several hits in an area of unallocated clusters identified by EnCase as a swap file.

Q: What is a swap file?

A: A swap file is a random area on a hard disk used by the computer's operating system to temporarily store data as a means to manage the available operating memory of a computer. The operating system will swap information as needed between the memory chips and the hard disk in order to process that information. As a result, temporary data is placed on the computer that cannot be viewed without special software designed for that purpose.

Q: What type of data is typically written to the swap file?

A: Any data that appears on the computer screen, even in the form of an unsaved word processing document or a Web page being viewed by the user is often written to the swap file by the operating system.

Q: What did you do after you identified search hits for the keyword Jenny86 in the swap file area?

A: I retrieved the full text of the information contained in the swap file and printed it out.

Q: I'm now handing you what has been previously marked as exhibit 31 and ask if you can identify it?

A: Yes. This is the print-out I made of the data contained in the swap file where my keyword search registered hits for the keyword Jenny86.

Q: If you would, please read the text as it appears on this printout.

A: The text appears in transcript form and reads, "Welcome to Yahoo Young Teen Chat [full text is read]"

[Exhibit is introduced into evidence.]

F. Evidence Found in File Slack

Q: What else did you find in your examination of the Defendant's computer?

A: I conducted a separate text string search of the forensic image of the Defendant's hard drive. In our investigation, we received additional information that the Defendant had corresponded approximately one to two years ago to another individual on more than one occasion. That person has since been convicted of possession of child pornography and sexual assault on a minor. This person's name is John Doe, and he commonly went by the nickname Lolita's Man. We conducted a text string search with the keyword Lolita's Man and registered a hit in an area of data known as file slack, which contained remnants of a deleted file.

Q: What is file slack?

A: Data storage areas on a hard disk are segmented into clusters. All the data constituting a file may occupy an entire cluster, or the file data may not take up all of the space in the physical cluster. The space between the end of a file and the physical end of a cluster is called the file slack. After the point in the cluster where the file ends, there may be pre-existing bytes in a cluster that are remnants of previous files or folders.

(Note: A projected PowerPoint slide or other form of demonstrative graphic illustrating this issue would be effective at this part of the examination.)

Example of a Demonstrative Trial Graphic

Q: What did you do after you identified search hits for the keyword John Doe in the area of file slack?

A: I retrieved the full text of the remainder of the document contained in the file slack, and printed it out.

Q: Could you determine what kind of document the remnant text in file slack was a part of?

A: Based upon my observation of the format of the two remaining paragraphs in the document and the signature block at the end of the document, it appears that the text recovered from file slack was the remnants of a correspondence of some type.

Q: I'm now handing you what has been previously marked as exhibit 32, and ask if you can identify it?

A: Yes. This is the print-out I made of the data contained in the file slack area where my text search registered a hit for the text string search Lolita's Man.

Q: If you would, please read the text as it appears on this print-out.

A: [The text is read into the record]

(Note: Because oral testimony of the recovery of file slack may seem too abstract to the jury it is recommended that a full screen shot of EnCase in from the "File View"

with the highlighted text hit in file slack be projected in order to show the full context of the relevant text.)

Q: Showing what has been pre-marked as exhibit 33 on the projection screen, does this look familiar to you?

A: Yes, that is a screen shot of the File View of EnCase I created, showing the search hit for "Lolita's Man" in file slack.

Q: Part of the text on the screen is in red, while the text before it is in normal black font. Does the text coloring have any significance?

A: The black text is the active, or non-deleted file that occupies the point from the beginning of the cluster to the end of that file. The red text represents the file slack in the area from the end of the non-deleted file to the end of the cluster.

[Exhibits 32 and 33 are introduced into evidence.]

G. Evidence of Windows Metafiles Recovered from Unallocated Clusters

Q: What else did you find in your examination of the Defendant's computer?

A: As part of my routine practice, I recovered all Windows metafiles that were located on the hard drive.

Q: What are Windows metafiles?

A: When a user sends a command to print a file, the Windows operating system makes a copy of that file and sends the copy to the printer. After the file is sent to the printer, Windows deletes that file. Windows does not inform the user that the copy, or metafile, has been made, nor can the user usually detect the existence of the metafiles without special software.

Q: How did you recover the metafiles in this case?

A: The EnCase software has an automated function that locates all the metafiles residing in normally unseen areas on a hard drive, decodes them, and outputs them to a separate folder allowing them to be viewed.

Q: What did you do after you utilized this software function that located the metafiles and outputted them to a folder?

A: I opened the folder and viewed each of the recovered metafiles.

Q: What did you find?

A: I found a text document in an e-mail format addressed to the Defendant's e-mail account. According to the e-mail header information, the message was sent from the account of Jenny86@hotmail.com.

Q: What does the fact that this e-mail document existed in the form of a metafile mean to you?

A: This recovered metafile means that this e-mail message was printed out from the Defendant's computer.

Q: I'm now handing you what has been previously marked as exhibit 34, and ask if you can identify it?

A: Yes. This is the printout I made of the metafile of the e-mail document from Jenny86@hotmail.com to the e-mail account of the Defendant.

Q: If you would, please read the text as it appears on this printout.